

Digital Twin Workshop – Student Guide

This guide supports the Digital Twin workshop and is designed to help you follow along, reflect on what we are building, and understand how each step contributes to a real digital twin workflow.

0. Creating a Bentley Education account and accessing workshop assets

Before we start building anything, everyone needs to be working from the same starting point. First, create a free Bentley Education student account. Once registered, create your [Cesium ion account](#) by using 3rd party authentication with your Bentley Education account. Once logged in, you will see the Cesium ion dashboard.



The image shows the Cesium ion sign-in interface. At the top is the Cesium ion logo. Below it is the heading "Sign in". Underneath is the section "Third-party authentication" which contains five buttons: Bentley (highlighted with a red border), Epic, GitHub, Google, and Sketchfab. Below these buttons is a horizontal line with the word "OR" in the center. Underneath the line is the label "Username or email" followed by a text input field. At the bottom left is a "Next" button, and at the bottom right is a link "Forgot password?". At the very bottom is a link "New to Cesium ion? Sign up for an account".

 **CESIUM** ion™

Sign in

Third-party authentication

 Bentley

 Epic

 GitHub

 Google

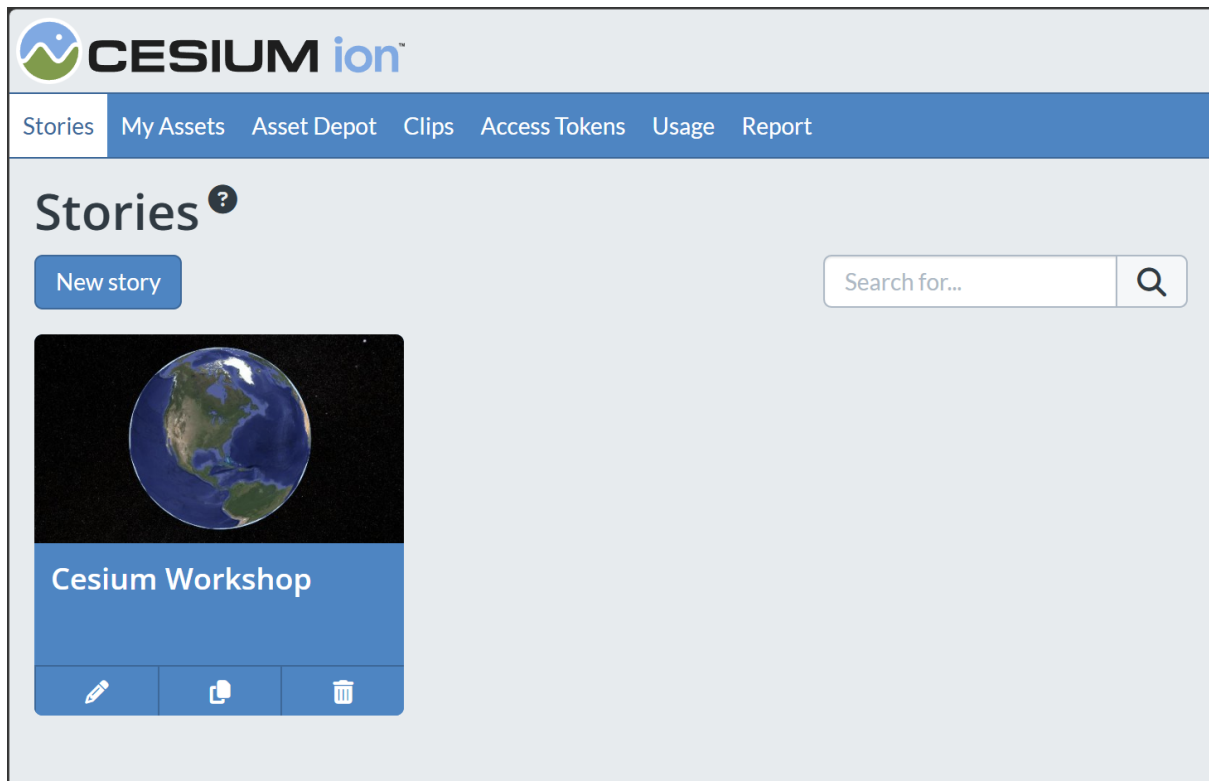
 Sketchfab

OR

Username or email

[Next](#) [Forgot password?](#)

New to Cesium ion? [Sign up for an account](#)



This dashboard is where you will access the tools used in today's workshop, including Cesium Stories, Cesium Sandcastle, assets, and access tokens.

***No local installation is required

To access assets that will be demonstrated during the workshop, head over to this [GitHub portfolio](#). These include building models, boundaries, and pre-tiled photogrammetry.

1. Workshop mindset

You are not expected to memorise code or understand every line you see today. The goal is to understand patterns, not syntax.

As you work through the labs, keep coming back to these three questions:

- What data is being used?
- What question does it help answer?
- How could this scale up to something bigger?

2. What you are building today

Throughout the workshop we will incrementally build a simple digital twin. Not a finished product, but a realistic foundation. By the end, your digital twin will include:

- Real-world terrain and global context
- Existing buildings and captured reality
- A proposed design model
- Tabular data connected to geometry
- Basic interaction and controls



3. Cesium Sandcastle

Cesium Sandcastle is a live coding environment for CesiumJS. It runs real production code directly in the browser. Why we use Sandcastle in this workshop:

- No installation or setup
- Instant feedback
- Same code used in real applications
- Safe environment for experimentation

Sandcastle is not the final destination, but the skills and patterns you learn here transfer directly to real projects.

4. Cesium Ion assets (quick reference)

In Cesium ion, everything is an asset. An asset is cloud-hosted, optimised, and streamable data.

Common asset types you will see today:

- Terrain and imagery
- 3D Tiles (buildings, photogrammetry, BIM models)
- Vector data such as GeoJSON

Once uploaded, assets are referenced by ID and can be reused across viewers, apps, and stories.



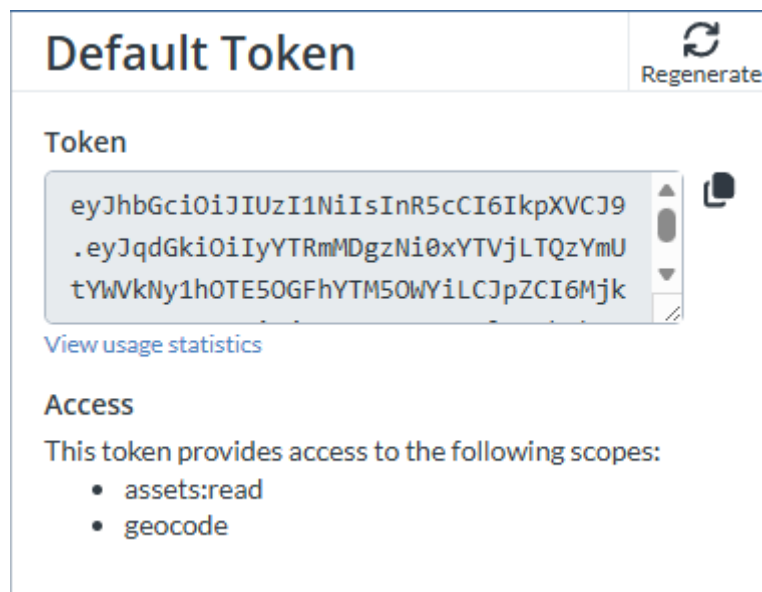
5. Access tokens (quick reference)

Access tokens authenticate your application with Cesium ion.

Key ideas to remember:

- Tokens control access, not the data itself
- Tokens have scopes (permissions)
- Best practice is one token per application

In this workshop, tokens are provided so you can focus on learning, not configuration. They can be found under the 'Access Tokens' tab in Cesium Ion.



6. Prompt engineering (vibecoding)

Alongside Cesium, you will use Large Language Models (LLMs) to help you build faster.

This works best when you give the model clear guardrails:

- Describe what you want to build
- Specify constraints (Sandcastle, CesiumJS)
- Be clear about data sources and goals

Prompt engineering is simply learning how to communicate intent clearly.

7. Lab structure

The 2 labs are structured to mirror how real projects evolve over time:

Lab 1.1 – Establishing context (area of interest, existing buildings)

Lab 1.2 – Captured reality (photogrammetry)

Lab 2.1 – Establishing context (area of interest, existing buildings)

Lab 2.2 – Google photorealistic 3D tiles

Lab 2.3 – Connecting data (turning models into insight)

Each lab layers new information on top of what already exists.

8. Prompt cheat sheet

A prompt cheat sheet is provided alongside this guide. Use it to:

- Set context for the LLM
- Ask for CesiumJS/Sandcastle-compatible code
- Keep generated code modular and understandable

What success looks like

Success today is not finishing every task.

Success is understanding how terrain, assets, data, and interaction come together to form a digital twin — and knowing how to grow it further.